

Competitiveness in American Seafood Requires Promoting Climate-Resilient Fisheries

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Abstract

Recent federal executive actions prioritize restoring American seafood competitiveness through deregulation and production targets, yet these policies overlook the fundamental threat posed by climate-driven natural hazards. We argue that true seafood competitiveness cannot be achieved through production gains alone because competitiveness also relies on stable production. Disaster resilience is therefore essential to long-term seafood competitiveness as it promotes stability and shortens recovery times. To address this policy gap, we propose a three-pillar framework that encompasses adaptive governance, a politically insulated safety net, and quantified climate risk assessments as the foundation for a durable, competitive, and resilient American seafood industry. We illustrate our point using fisheries from U.S. Gulf states as an example of a system increasingly exposed to natural hazards, the most prominent of which are hurricanes. This climate vulnerability is not unique to the Gulf nor to hurricanes. Other climate-driven impacts such as marine heatwaves and harmful algal blooms are becoming increasingly prevalent and impose analogous shocks along U.S. waters.

Keywords: climate adaptation, disaster resilience, fisheries management, Gulf of Mexico, hurricane impacts, seafood policy

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1 Resilience is the foundation of seafood competitiveness

Recent executive actions and federal directives in the United States have prioritized restoring American seafood competitiveness to ensure food security and economic growth [Executive Office of the President of the United States, 2025]. However, policies do not exist in a vacuum. Hazards such as hurricanes, harmful algal blooms, and marine heatwaves may inadvertently undermine these goals. As climate change raises ocean and atmospheric temperatures, the frequency and intensity of these shocks is increasing [Oliver et al., 2019, Plecha and Soares, 2020, Vecchi et al., 2021], placing the very regions essential to domestic seafood production in the crosshairs. Here, we analyze the intersection of seafood policy and natural disaster management using American states along the Gulf of Mexico (currently referred to as the Gulf of America by the U.S. government) as an example of a region increasingly exposed to natural hazards, particularly in the form of hurricanes (Fig.1). We argue that true competitiveness cannot be achieved without a comprehensive resilience strategy for the region. Importantly, our insights and arguments extend to other regions in the United States where fisheries are vulnerable to other natural hazards and climate impacts.

The Gulf of Mexico is the backbone of the southern coastal economy, yet it faces both the highest and fastest-growing exposure to hurricanes in the nation ([FEMA, 2025a]; Fig. 1A). Encompassing wild-capture commercial and recreational fisheries across Texas, Louisiana, Mississippi, Alabama, and Florida, the region generates approximately \$910 million annually in commercial wild-capture dockside revenue (roughly 15% of the U.S. wild-capture total) and lands the highest number of commercially harvested shrimp and oysters in the nation [NOAA Fisheries, 2024b]. The Gulf’s recreational fishery is equally important, accounting for over one-third of the nation’s fishing trip-related expenditures at \$5.1 billion [NOAA Fisheries, 2024a]. Together, these commercial and recreational sectors constitute a major share of the U.S. seafood economy, supporting over 250,000 full- and part-time jobs [NOAA

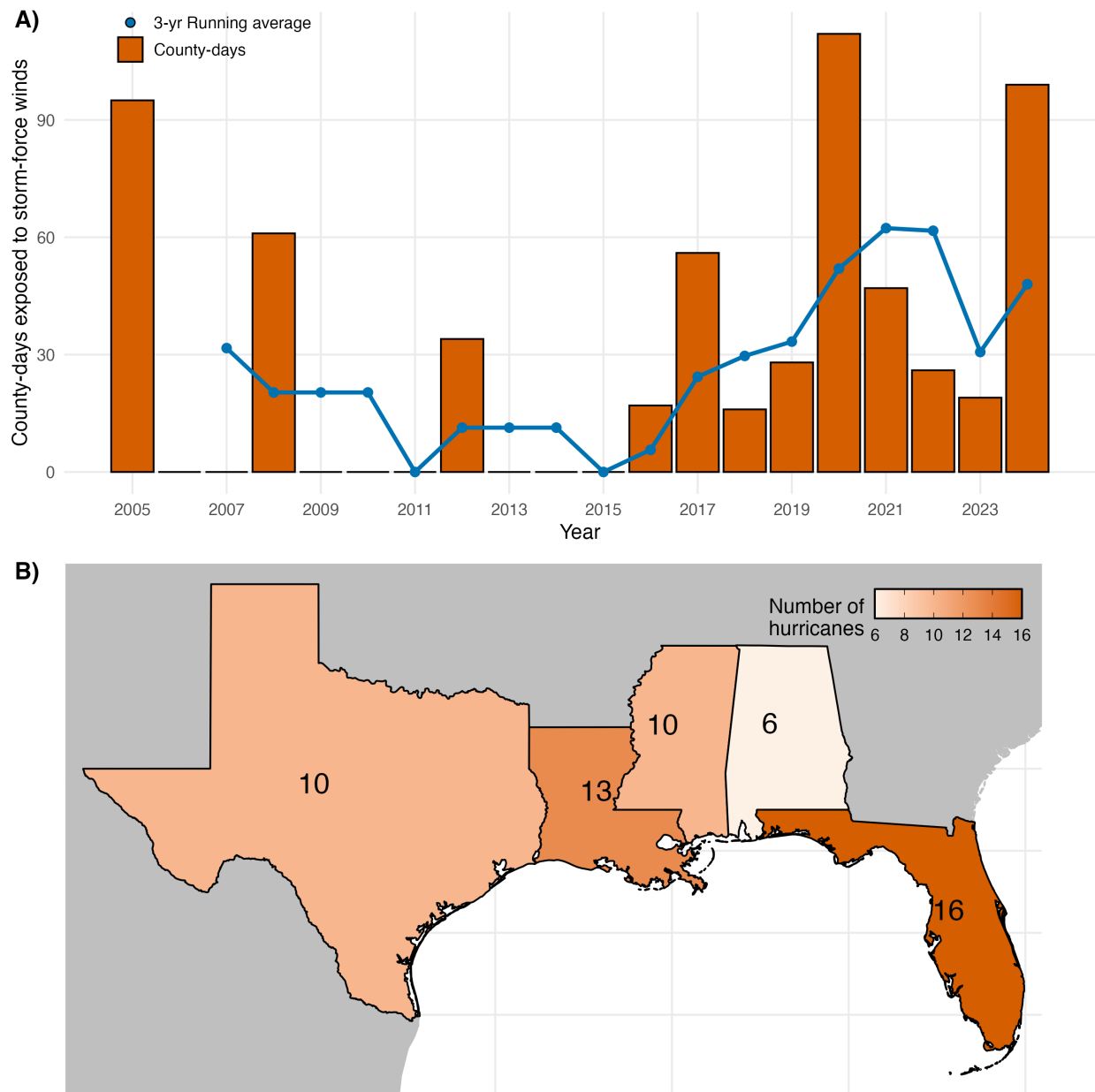


Figure 1: **Fisheries in Gulf states are increasingly exposed to hurricanes.** In panel **A)**, bars show annual hurricane-season (June–November) county-days exposed to maximum sustained winds of at least 17.5 m/s (tropical-storm force) across counties with known fishery production and within 50 km of the coast in Texas, Louisiana, Mississippi, Alabama, and Florida. The line is a 3-year running average. In panel **B)**, colors and number insets in the map indicate historical (2005-2024) State-level exposure to tropical storm-force winds.

52 Fisheries, 2024a], while sustaining centuries-old cultural identities and critical food systems
 53 (Tab. 1). It follows, then, that the most productive and valuable fisheries in the Southeastern
 54 United States are concentrated in its most storm-prone waters (Fig. 1B).

Table 1: Commercial and Recreational fisheries in Gulf States are an important source of livelihoods and food.

State	Employment		Production		Income		Sales		Value Added	
	Com.	Rec.	Com.	Rec.	Com.	Rec.	Com.	Rec.	Com.	Rec.
Florida	121,710	28,546	49,902	60,556	4,579	1,350	24,596	3,936	8,207	2,401
Texas	41,171	5,605	27,889		1,259	261	5,584	793	2,049	450
Louisiana	32,514	4,552	413,855		623	171	1,638	590	838	294
Alabama	6,971	5,506	13,871	8,685	140	217	443	745	196	399
Mississippi	6,954	324	141,018	4,917	127	13	329	72	165	30

Data come from [NOAA Fisheries, 2024a]. Each economic metric is split by fishing sector: Com. = commercial fisheries; Rec. = recreational fisheries. Units: Employment is full- and part-time jobs; Production is in metric tons; Income, Sales and Value Added are in millions of U.S. dollars. Empty cells indicate that the metric is not separately reported for that state-sector combination.

In renewable resource-driven industries like fisheries, long-term competitiveness requires more than just maximizing fishery yields; it demands resilience. We define resilience as the seafood sector’s ability to maintain stable, continuous production systems capable of reliably supplying markets year over year (*sensu* Villaseñor Derbez et al. [2025]). Climate-driven shocks directly undermine this stability by damaging infrastructure, disrupting supply chains, increasing operating costs, and prolonging recovery periods. Because reactive disaster relief only addresses the aftermath, proactive resilience planning is critical to minimizing and absorbing the initial impact of these disruptions.

Hurricanes and storms impose a double jeopardy on Gulf fisheries by simultaneously degrading the ecosystem and destroying physical capital. For example, storms induce rapid salinity fluctuations, turbidity, and physical damage to habitats like seagrass and oyster beds, which in turn destabilize food webs and reduce biomass of heavily modified environments [Conner et al., 1989, Monnereau et al., 2015, Rothenberger et al., 2018]. In addition, storm surge and wind destroy critical coastal infrastructure such as marinas, icehouses, and processing facilities, while fishers can be left without homes, vessels, or equipment [Caffey et al., 2019, Impact Assessment Inc, 2007]. In the aftermath, the lack of electricity, clean water, and fuel, combined with crew members focusing on personal and household recovery, further constrains fishers’ ability to resume production promptly [Ingles, 2007, Impact Assessment

Inc, 2007, Caffey et al., 2019].

Fishery production losses from these disruptions are costly, with existing estimates ranging from \$100,000 to \$100 million per storm [Caffey et al., 2007, Solís et al., 2013]. Storm-event damages to commercial seafood infrastructure increase this toll further, with estimates suggesting additional losses up to \$260,000 per storm in affected communities [Erlambang, 2008].

Beyond immediate losses, the erosion of productive capacity damages broader seafood competitiveness when financial constraints stall community recovery. High insurance premiums leave fishers to absorb the full cost of rebuilding out-of-pocket, while limited access to credit—stemming from poor credit histories and Gulf fleet over-capitalization—restricts their ability to recapitalize [Caffey et al., 2019, Impact Assessment Inc, 2007]. Compounded by fuel supply shocks and shrinking profit margins, these financial barriers delay the resumption of harvest and processing activities, multiplying total economic losses and amplifying market volatility.

Given these multiple layers of vulnerability, viewing financial recovery merely as a social safety net ignores its primary economic function: rapid capital deployment minimizes output losses and allows fisheries to reclaim their market position sooner. Disaster resilience should therefore be understood not simply as emergency preparedness, but as an economic asset that preserves productive capacity and prevents climate shocks from neutralizing or even eroding competitive advantages. Yet, despite the centrality of resilience to maintaining seafood competitiveness, current U.S. policy approaches do not fully integrate this relationship into their design.

2 How current U.S. seafood policy fails to foster resilience

A complex web of institutions mitigate hurricane impacts in the United States. Disaster response operates through a multi-level governance system in which municipal and county governments provide the initial emergency response, state agencies coordinate regional assistance, and federal agencies provide support when damages exceed state and local capacity (See Box 1 for details). Federal support is bifurcated: NOAA focuses on forecasting and damage assessment, while FEMA and the Small Business Administration coordinate broader financial recovery through loans and disaster assistance [FEMA, 2025b]. State agencies, such as the Florida Fish and Wildlife Conservation Commission and the Louisiana Department of Wildlife and Fisheries, may bridge local gaps with emergency closures and state-level aid.

Even with this apparatus in place, the current policy landscape falls short of supporting resilience in four connected ways. First, disaster relief remains fundamentally reactive, fragmented, and slow to deploy. Fishers frequently note that state and federal assistance is hindered by inadequate funding, complex administrative procedures, and a disconnect from local needs [Ingles, 2007]. Consequently, the sector leans heavily on familial networks, community ties, and non-governmental organizations (NGOs) for immediate post-disaster recovery, including boat repairs, crew replacement, and emergency financial aid [Dunning, 2020, Karlsson and McLean, 2020, Ramenzoni et al., 2020]. Yet, while these informal safety nets provide vital short-term relief, they are not a substitute for systemic economic resilience. The existing recovery framework is administratively burdensome and backward-looking, which highlights the urgent need for proactive policies that are pre-funded, rules-based, and explicitly designed to sustain productive capacity before and after major shocks.

Box 1 - Federal and State Support for Disaster and Emergency Assistance

Federal disaster assistance in general is authorized under the Robert T. Stafford Disaster Relief and Emergency Assistance Act, which allows governors to request presidential emergency or major disaster declarations following severe events such as hurricanes [Federal Emergency Management Agency (FEMA), 2025a]. Once approved, agencies coordinate through the National Response Framework and the Incident Command System, which organize preparedness, emergency response, damage assessment, and long-term recovery efforts across federal, state, and local governments [Federal Emergency Management Agency (FEMA), 2025b].

Within the fisheries sector, the primary federal agencies involved in hurricane preparedness, response, and recovery are NOAA, NOAA Fisheries, and the Federal Emergency Management Agency (FEMA). NOAA's National Weather Service and National Hurricane Center provide forecasting and warnings that guide evacuation and preparedness decisions prior to landfall, while FEMA coordinates federal emergency response and recovery under the National Response Framework [NOAA, 2025, NWS, n.d.]. NOAA Fisheries conducts post-disaster fisheries impact assessments, habitat evaluations, and fishery disaster determinations under the Fishery Resource Disasters Improvement Act [NOAA Fisheries, 2025]. Because NOAA Fisheries also manages marine resources under the Magnuson–Stevens Fishery Conservation and Management Act, disaster recovery efforts frequently intersect with fisheries management responsibilities. Hurricanes routinely disrupt stock assessments, damage habitat, delay scientific surveys, and require temporary fisheries closures or emergency regulatory adjustments coordinated between NOAA Fisheries, Regional Fishery Management Councils, and state agencies [Louisiana Department of Wildlife and Fisheries, 2025, Florida Fish and Wildlife Conservation Commission, 2024, Texas Parks and Wildlife Department, 2025].

Following hurricanes, Congress may appropriate fishery disaster relief administered by NOAA Fisheries, while FEMA coordinates broader assistance for individuals, businesses, and governments. Additional recovery support may come from Small Business Administration Disaster Loans, Economic Injury Disaster Loans, Disaster Unemployment Assistance, and the Community Development Block Grant–Disaster Recovery program, which supports long-term rebuilding in coastal communities [FEMA, 2025b].

State agencies in hurricane-prone regions also play a significant role in preparedness and recovery. Agencies such as the Florida Fish and Wildlife Conservation Commission, the Louisiana Department of Wildlife and Fisheries, and the Texas Parks and Wildlife Department oversee emergency fisheries closures, assess damages, coordinate with federal agencies, and distribute state-level assistance [Florida Fish and Wildlife Conservation Commission, 2024, Louisiana Department of Wildlife and Fisheries, 2025, Texas Parks and Wildlife Department, 2025]. Sea Grant programs across Gulf states further support coastal communities through preparedness planning, continuity training, and recovery workshops [Texas Sea Grant College Program, 2025, Florida Sea Grant, 2025].

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120 Second, the 2025 Executive Order on Restoring American Seafood Competitiveness empha-
121 sizes reducing regulatory barriers in ways that risk undoing conservation and recovery, thereby
122 weakening long-term adaptive capacity in pursuit of short-term output. Its implementation
123 has primarily focused on technology adoption and streamlining NOAA's management to
124 boost production and reduce the \$20 billion seafood trade deficit. Although this mandate
125 calls for modernizing data collection and making management more responsive to real-time
126 conditions(including environmental shocks), we argue that its core emphasis on deregulation

and bypassing conservation efforts does not prioritize resilience and may even weaken it.

Third, by prioritizing output and deregulation, current policy overlooks the factors needed to prevent and recover from climate-driven shocks—a vulnerability that is not unique to hurricanes. Marine heatwaves and harmful algal blooms impose analogous shocks: in the Pacific and Atlantic coasts, for instance, marine heatwaves have caused temporary redistribution of target species, leading to 40-60% reductions in biomass and catch [Chaikin et al., 2026, Free et al., 2023, Villaseñor Derbez et al., 2024]. In extreme cases, marine heatwaves can also induce total collapse of populations, as observed for snow crab in the Bering Sea during 2021 [Szuwalski et al., 2023]. Harmful algal blooms also pose a serious threat to fisheries because high concentrations of toxic micro-algae can trigger fishery closures, leading to large economic losses [Jardine et al., 2020, Mao and Jardine, 2020]. Critically, fisheries that are more resilient to climate shocks such as marine heatwaves have also been shown to be more resilient to market shocks such as those brought about by the COVID-19 pandemic, which suggests that climate resilience confers broad cross-domain benefits [Villaseñor Derbez et al., 2025].

Fourth, the broader erosion of federal scientific capacity undermines the forecasting and assessment needed to prepare proactively. For example, all the environmental shocks mentioned above are climate- and weather-related events with some degree of predictability [McGowan et al., 2017, Arcodia et al., 2025]. Accurate forecasts have been shown to be widely effective at reducing associated damages by enabling adaptation [Anand, 2025, Shrader et al., 2026, Molina and Rudik, forthcoming]. Therefore, any viable strategy for proactive fisheries management must prioritize investments in the scientific programs that monitor and forecast these recurrent climate shocks [Trainer et al., 2020].

Despite its strategic importance, scientific capacity is increasingly vulnerable. The recent suspension and restructuring of the Sixth National Climate Assessment (NCA) and removal of federal climate data repositories represent a withdrawal of information essential to understanding environmental variability. Without timely assessments provided by the NCA,

154 responses to climate shocks are likely to remain reactive.

155 Compounding this vulnerability, staffing and budget pressures across NOAA and the
156 National Marine Fisheries Service (NMFS) are actively eroding the capacity of federal
157 observing systems. The U.S. Government Accountability Office has documented persistent
158 workforce constraints within NOAA that compromise core functions, including environmental
159 monitoring and fisheries stock assessments [U.S. Government Accountability Office, 2022,
160 2023]. This resource strain extends to the National Weather Service, where observational tools
161 like upper-air weather balloons serve as critical inputs for severe weather models. Without
162 these robust data streams, forecast accuracy during high-impact events degrades significantly
163 [NWS, n.d., National Oceanic and Atmospheric Administration, National Weather Service,
164 2020]. As a result, shrinking scientific and forecasting capacity leaves fishing communities
165 with less time and reliable information to anticipate storms, directly increasing their exposure
166 to disruption.

167 Finally, NOAA’s proposed 2026 framework to narrow NMFS management and science
168 through a Risk/Value Matrix threatens to further constrain adaptive capacity. Although the
169 framework is intended to prioritize limited resources, the Regional Fisheries Management
170 Council Coordination Committee has warned that narrowing assessment and monitoring
171 priorities may severely weaken the scientific foresight needed to maintain stable domestic har-
172 vest under increasingly variable environmental conditions [Council Coordination Committee,
173 2026]. By reducing flexibility in region-specific management systems, this framework further
174 undermines the broader national objective of strengthening domestic seafood production.

175 Overall, the current policy landscape attempts to optimize the sector’s output in pursuit
176 of a competitive advantage, yet it does so without systematically addressing the increasing
177 threat of climate-driven shocks, including hurricanes, which continue to undermine that
178 output. This leaves Gulf state fisheries—and national fisheries in general—in a delicate state
179 where short-term gains in efficiency can be quickly erased by a single catastrophic event.

180 Achieving the federal mandate of seafood competitiveness demands strengthening FEMA

and NOAA’s capacity to predict and mitigate the impacts of climate shocks. This capacity must be recognized as an act of industrial policy rather than simple disaster relief. To overcome current vulnerabilities, disaster preparedness must be treated not as a reactive safety net, but as a strategic investment that stabilizes production. By acknowledging resilience as a strict prerequisite for market stability, the United States can ensure the Gulf and other at-risk regions remain competitive pillars of the national seafood economy. Making this shift ultimately requires decision-support science that traces how climate shocks propagate across interlinked supply chain components, rather than analyzing vulnerabilities in isolation.

Moving beyond this reactive framework and treating ecological monitoring and economic impact assessment as essential public capital offers a path toward proactive management strategies that enhance resilience and competitiveness in the U.S. seafood sector. Crucially, these arguments build on, rather than displace, existing efforts. For example, the South Atlantic Fishery Management Council’s East Coast Climate Change Scenario Planning initiative already demonstrates the integration of climate foresight into regional fisheries governance. While this initiative successfully identifies vulnerabilities and generates a suite of possible responses, implementation ultimately depends on voluntary uptake by individual management bodies [National Oceanic and Atmospheric Administration Fisheries (NOAA Fisheries), 2025]. Our proposed approach aligns with these programs, and extends them by tying foresight to enforceable governance, a pre-funded safety net, and quantified economic risk assessment.

3 Three Pillars to Support Seafood Competitiveness

We argue that securing the competitiveness of the American seafood sector requires a multi-tiered policy approach. This approach is particularly relevant for fisheries in the Gulf states, which are acutely exposed to environmental shocks such as hurricanes. While our

focus is on commercial and recreational fisheries, we recognize that the resulting disruptions extend into processing, distribution, and retail. Within that scope, the framework is organized around three pillars: (1) robust and adaptive governance; (2) an independent safety net; and (3) enhanced clarity on systemic climate risk. The framework is intended to be integrated rather than modular: governance shapes exposure to shocks, the safety net influences recovery speed, and climate-risk assessment improves anticipatory decision-making across the system.

Pillar 1: Robust and adaptive governance. Clear and adaptive rules are essential to supporting efficient and climate-resilient fisheries, which in turn provide the foundation for market competitiveness. State and federal rules govern who has access to natural resources, as well as how much and when they can extract. When designed properly, these rules can ensure well-functioning fisheries, prevent collapse, and support the rebuilding of fish stocks [Costello and Deacon, 2007, Costello et al., 2008, Young et al., 2018, Frank and Oremus, 2025]. Clarity reduces management complexity and facilitates timely guidance from managers and disaster relief agencies, while adaptability allows fishers and managers to respond to dynamic climate conditions and better withstand environmental shocks. For example, fishers may adapt by modifying their gear or adjusting the timing and location of fishing [Ilosvay et al., 2022, Mason et al., 2022, Welch et al., 2025], while managers can adjust spatial or temporal closures, issue temporary permits to target alternative species, or modify—both downward and upward—total allowable catch limits [Mason et al., 2022]. More broadly, adaptability implies avoiding “one-size-fits-all” approaches that fail to account for the diversity of interactions in fisheries as complex social-ecological systems [Mahon et al., 2008].

Along this dimension, the United States already has a robust and adaptive approach to fisheries management anchored in the Magnuson–Stevens Fishery Conservation and Management Act [Frank and Oremus, 2025]. However, the stability of fisheries production may be threatened by ongoing executive actions incorrectly arguing that current levels of fisheries production are constrained by excessive regulation. There is ample evidence showing

that production is primarily constrained by economic incentives, or lack thereof, rather than by regulatory constraints [Oremus et al., 2023].

Efforts to enhance American seafood competitiveness should focus on reinforcing clear and adaptive rules, whether already in place or still to be developed, and on simplifying existing frameworks without compromising future harvests or catch stability in pursuit of short-term increases in yield. In catch-share contexts in particular, the gains from adaptive rules depend heavily on pairing them with proper stock assessments, as well as secure and durable access rights [Arnason, 1996].

Pillar 2: Independent safety net. While regulations can help mitigate the impacts of environmental shocks, they cannot prevent them altogether. As such, an independent safety net can enhance resilience when such shocks inevitably occur. To achieve this goal, the sector requires a politically insulated network of financial and logistical support mechanisms. This includes a) expanding access to affordable index-based risk insurance (also known as parametric insurance); and b) streamlining Community Development Block Grant–Disaster Recovery funds.

Index-based insurance models are designed to pay out automatically when a pre-defined trigger, such as a wind speed threshold or storm surge height, is exceeded. This mechanism delivers funds faster than case-by-case federal aid programs, which rely on lengthy evaluation and deliberation processes that ultimately delay recovery [Ingles, 2007]. While these models have long been used in agriculture to buffer the impacts of drought and reduce social conflict over shared resources [Tadesse et al., 2015, Gehring and Schaudt, 2026], they have only recently gained traction in the marine realm to hedge against climate-related risks. One example is the Philippines, which pioneered their use through an index-based microinsurance program against exposure to typhoons for small-scale fishers [Rare and Willis Towers Watson, 2025, UNEP Finance Initiative, 2023].

Another example comes from the Caribbean, where several nations now employ index-

based insurance via COAST (Caribbean Oceans and Aquaculture Sustainability Facility), a program developed jointly by the Caribbean Catastrophe Risk Insurance Facility and the World Bank with U.S. State Department funding. Launched in 2019 and piloted in Grenada and Saint Lucia [Sainsbury et al., 2019], the program has since expanded to Anguilla, Barbados, Saint Vincent and the Grenadines, and Turks and Caicos. Although COAST insures sovereign nations rather than individual fishers, it successfully generated payouts for Grenada following Hurricane Beryl in 2024 [CCRIF SPC, 2025], providing a proven template for the design of similar systems. In practice, a U.S. equivalent would likely combine private or mutual insurance provision with public support for the underlying data infrastructure, federal backstopping, and subsidies to ensure affordability in chronically high-risk communities.

Future efforts should also prioritize rebuilding grants that reduce the sector’s exposure to hazards (e.g., storm-resistant cold storage and fuel systems). To avoid creating incentives analogous to harmful fishing subsidies (e.g., [Englander et al., 2023]), these mechanisms should be tied to preserving or restoring productive capacity and continuity of operations rather than expanding fleet capacity [Molina, 2022]. By replacing *ad hoc* supplemental appropriations with pre-priced, rules-based transfers tied to existing capacity, this design also lowers expected federal disaster outlays and helps contain the climate-induced fiscal risk documented in the Fifth National Climate Assessment [Hsiang et al., 2023]. To be effective, we *emphasize* that these mechanisms must provide reliable protection against environmental risk and remain insulated from political disruption.

Pillar 3: Enhanced clarity on systemic climate risk. Finally, we argue that the sector requires a clear understanding of, and effective communication about, systemic climate risk. A central component of this pillar relies on systematically learning from past impacts while also stress-testing the sector against novel combinations of future climate conditions [Pershing et al., 2019, Dillon et al., 2014]. Policymakers should prioritize the development and dissemination of science-driven, quantitative risk assessments of hurricanes and other climate-

related hazards, including their economic consequences [Link et al., 2015, Parmesan et al., 2022]. These assessments should be comprehensive, from stock health and fleet operations to infrastructure and distribution networks, and be paired with the development of a transparent menu of mitigation and adaptation options tailored to fisheries and coastal communities.

Together, these three pillars form an integrated framework for building a competitive and resilient American seafood industry. To ensure the framework translates from policy theory to measurable outcomes, it must be accompanied by indicator-based monitoring at the federal and state levels. Example indicators for Pillar 1 include the time elapsed between a declared disaster and the corresponding emergency regulatory action, and the share of fisheries with pre-specified adaptive rules in their management plans. For Pillar 2, indicators could include the median time-to-disbursement of post-disaster aid and the share of at-risk landed value covered by parametric instruments. Finally, potential indicators for Pillar 3 include the existence and coverage of quantified, regionally explicit climate-risk assessments tied to each major fishery. By reinforcing governance, stabilizing risk exposure, and improving information on climate threats, short-term efficiency gains can be sustained even as climate risks intensify.

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Data Availability Statement

All data and code to produce Fig. 1 are available at https://github.com/jcvdav/hurricanes_resilient_fisheries.

Conflict of Interest Statement

The authors declare no conflicts of interest.

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